RISHI THAKUR

AI SOFTWARE DEVELOPMENT ENGINEER

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PROFESSIONAL SUMMARY

AI Software Development Engineer (M.Tech CSE, GATE CS qualified) experienced in designing, productionizing, and evaluating machine learning systems and deep learning models. Proficient in PyTorch, Python backend development, REST API integration, and high-throughput data pipelines. Skilled in handling class imbalances, optimizing inference latency, and deploying scalable data models.

SKILLS

- **AI Engineering & Deep Learning:** PyTorch, TensorFlow, Transformers, Hugging Face, BERT Embeddings, Diffusion Models (DDPM/DDIM), Recurrent Encoders (GRU, Bi-LSTM), Scikit-learn, XGBoost
- **Software Engineering & Deployment:** Python (Advanced OOP, Data Structures, Concurrency), RESTful APIs (FastAPI, Flask), Model Serving, Docker Basics, Git/GitHub, Linux/Bash, PyTest, CI/CD Basics
- **Data Infrastructure & Databases:** Advanced SQL (MySQL, MariaDB), Relational Schema Design, Data Pipelines, Resampling (SMOTE + Tomek Links), Latency Optimization, Star Schema
- **Communication & Spoken Skills:** Cross-Functional Engineering Collaboration, Technical Architecture Documentation, Model Explainability, Agile Sprints, API Design Reviews

WORK EXPERIENCE

Chhattisgarh Infotech Promotion Society (CHiPS) | Raipur, India | 06/2025 – 03/2026
AI & Software Engineering Intern • Dept. of Health & Family Welfare, Govt. Of Chhattisgarh

- Engineered automated data ingestion and preprocessing pipelines linking biometric attendance logs with State Treasury payroll records across **33 districts**.
- Built an operational **3-tier risk classification engine** in Python and SQL, identifying compliance gaps across bonded medical officers.
- Standardized data schemas and backend integration interfaces between MariaDB, State Treasury, and National Next-Gen e-Hospital systems.
- Optimized backend database queries and feature extraction routines, ensuring **100% data fidelity** across high-volume state databases.
- Collaborated with National Informatics Centre (NIC) teams to deploy analytical reporting tools and document technical system architectures.

KEY PROJECTS

Proactive 5G Handover Prediction (Diffusion Models) • *M.Tech Thesis — Edge AI Architecture & Telemetry Pipeline*
Technologies: Python, PyTorch, GRU, DDPM/DDIM, Scikit-learn, Random Forest, SMOTE, Tomek Links

- Architected an edge-optimized AI pipeline pairing a **GRU encoder** with a **Diffusion Model (DDPM/DDIM)** forecasting signal drops from **spatio-temporal** telemetry.
- Resolved **98%/2% class imbalance** using **SMOTE + Tomek Links**, deploying a two-stage Random Forest policy with custom loss to achieve a **97.64% F1-score**.

AI vs. Human Text Classification Pipeline • *Production NLP Pipeline & Deep Learning Systems*
Technologies: Python, BERT, XGBoost, Bi-LSTM, Transformers, 1D-CNN, TensorFlow, Scikit-learn

- Engineered dual NLP production pipelines: extracted 768-dim **BERT contextual embeddings** paired with **XGBoost**, and built a deep hybrid Bi-LSTM/Transformer service.
- Applied stratified sampling, NFKD text normalization, and early stopping, achieving **99% Precision** and **98% Recall** with low-latency inference.

EDUCATION

M.Tech in Computer Science & Engineering | IIIT Naya Raipur (CGPA: 7.90) | 08/2024 – 07/2026
B.Tech in Computer Science | Amity University Chhattisgarh (CGPA: 7.71) | 08/2019 – 06/2023

CERTIFICATIONS & KEY QUALIFICATIONS

- **Qualified GATE Computer Science (CS)**, 2024
- **Python for Data Science & AI** (Coursera)
- **SQL & Database Design** (Udemy)
- **Celonis Process Mining Foundations** (Credly)
- **Microsoft Power BI** (Udemy)
- **IELTS Academic:** 7.5 Band Score